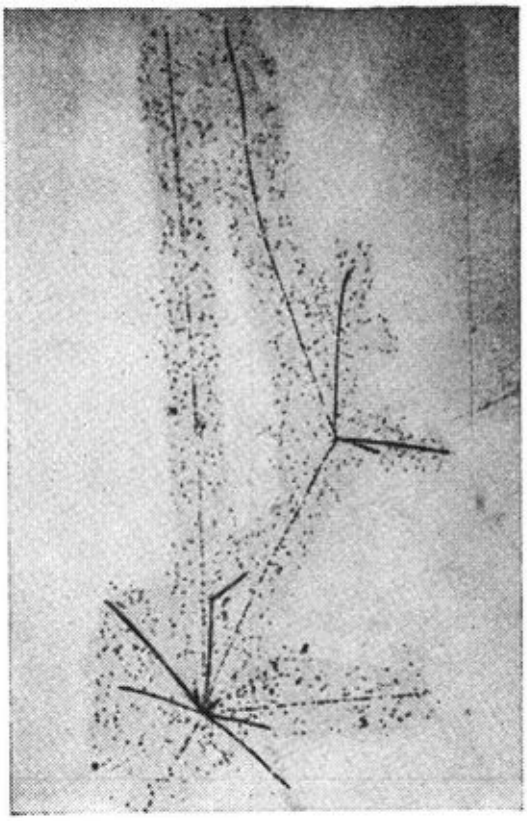


Figure 25B.—Primary and secondary nuclear disintegration.

A last, quite especially important case of the disintegration of nuclei is *nuclear fission* (the splitting of nuclei), discovered by Hahn and Strassmann in Berlin in 1938. This is what occurs in this case: It may happen that a heated nucleus ejects no individual particles at first, but begins to vibrate as a whole, utilizing all or part of the introduced energy for the excitation of these vibrations. This is often the case when a uranium nucleus is bombarded by a neutron. A reaction ensues then, which is shown diagrammatically in Figure 26. The nucleus, orginally spherical in shape, will vibrate first so as to assume alternately an elongated and a flattened elliptical form. The longitudinal deformation may reach a degree where the nucleus—like an iron rod which is about to break in two—becomes very thin approximately in its middle, and finally breaks into two more or less equal parts. Splinters fly off, as a rule, since several neutrons are ejected.

The fact that such fission is possible at all and is most likely to occur in the heaviest nuclei, is easy to understand. The nuclear forces which guarantee the stability of a nucleus are opposed by the electric force of repulsion which increases with the nuclear mass, since, considered on the whole, the charge also increases with the mass. The repulsion produces a decrease of the binding energy per particle, and thus a decrease of the stability with the increase in the mass. If in addition to this factor, the stability of the nucleus is imperilled by its vibration, the electric force of repulsion is able to make its